M2 COMMUNITY NODE

Build Guide

v1.0 | April 2026

Build your own field-portable community communications node — ATAK · Matrix · Mesh · Monero

capableenough.org

System Capabilities

Capability	Technology	Internet?
Tactical maps & positions	OpenTAK Server + ATAK	**No**
Encrypted group chat	Matrix (Conduit) + Element Web	**No**
Push-to-talk voice	Mumble + Mumla app	**No**
Off-grid mesh radio	Reticulum + Meshtastic (LoRa)	**No**
Anonymous access	Tor hidden services + I2P	**No**
Clearnet remote access	Cloudflare Tunnel	*Yes*
Remote VPN	Headscale (self-hosted WireGuard)	*Yes*
DNS filtering + Android bypass	AdGuard Home	**No**
Financial sovereignty	Monero pruned node	**No**

Design Principles

- Privacy-first. Censorship-resistant. Open-source only.
- Field-deployable in under 15 minutes — wall power, vehicle 12V, or UPS.
- Commodity hardware only. No vendor lock-in. ~$994 recommended build.
- All core services run with zero internet — air-gap capable.
- Built on two Raspberry Pi 5 16GB boards. Easy to source, easy to replace.

System Architecture

```
  INTERNET / CELLULAR
         |
  [GL.iNet Slate AX]  192.168.8.1
  WiFi: CommunityNode / NodeAdmin
         |
  [TP-Link TL-SG108S Switch]
    |                   |
[Pi #1 Comms]     [Pi #2 Tactical]
192.168.8.10      192.168.8.20
256GB NVMe        1TB NVMe
    |                   |
Conduit (Matrix)   OpenTAK Server
Element Web        Mumble
Nginx              Reticulum
Tor + I2P          NomadNet
Cloudflared        Monerod
AdGuard Home       Headscale
                   [LEFT LoRa: RNode / Reticulum]
                   [RIGHT LoRa: Meshtastic]
```

Node Roles

	Pi #1 — Comms	Pi #2 — Tactical
IP	192.168.8.10	192.168.8.20
Storage	256GB M.2 2230 NVMe	1TB M.2 2230 NVMe
Key services	Matrix, Element, Nginx, Tor, I2P, Cloudflared, AdGuard	OpenTAK Server, Mumble, Reticulum, Monerod, Headscale
Radio	*None*	2× Heltec V3 LoRa (USB)
Runtime mode	Docker Compose	OpenTAK = native systemd; others = Docker

Rack Layout

```
U8  Custom 1U LoRa panel  [LEFT: RNode]  [RIGHT: Meshtastic]
U7  GeeekPi 2U Touchscreen (node status display)
U6  (touchscreen continues)
U5  GL.iNet Slate AX + TP-Link Switch (shared 1U shelf)
U4  GeeekPi 1U Dual Pi 5 Mount  [Pi #1 | Pi #2]  (NVMe built in)
U3  Open
U2  Tupavco PDU (rear rails, outlets face inward)
U1  Tripp Lite BC600R UPS (rack floor)
```

Bill of Materials

All prices are approximate (verified Feb–Apr 2026). Verify before ordering — Pi 5 and M.2 2230 prices fluctuate.

Compute & Storage

Item	Qty	Est. Price	Where
Raspberry Pi 5 16GB	2	~$160 ea	*Amazon/PiShop.us*
Pi 5 Official Active Cooler	2	~$11 ea	*PiShop.us*
GeeekPi 1U Dual Pi5 Mount (B0F7XBVV4D)	1	~$60	*Amazon*
WD SN740 256GB M.2 2230 NVMe — Pi #1	1	~$40	*Amazon (B0C6MVP42M)*
Crucial P310 1TB M.2 2230 NVMe — Pi #2	1	~$130	*Amazon/Newegg*
SanDisk Extreme 32GB microSD ×2	2	~$12 ea	*Amazon*

Rack & Networking

Item	Qty	Est. Price	Where
GeeekPi 8U 10” Mini Rack (RackMate T1)	1	~$95	*Amazon*
GeeekPi 2U Touchscreen Display	1	~$70	*Amazon*
GL.iNet Slate AX (AXT1800)	1	~$120	*GL.iNet store*
TP-Link TL-SG108S 8-Port Switch	1	~$27	*Amazon/CompSource*
1U Shelf (for switch + GL.iNet)	1	~$18	*Amazon*
Short Cat6 patch cables (6”–1’), 5-pack	1	~$12	*Amazon*

Power

Item	Qty	Est. Price	Where
Anker 747 GaNPrime 150W Charger	1	~$65	*Amazon (B09W2PNLX7)*
Tupavco 10” 1U Rack PDU	1	~$35	*Amazon*
Tripp Lite BC600R UPS (600VA/300W)	1	~$70	*DigiKey / Amazon*
USB-C to USB-C cables 1’ 3-pack	1	~$10	*Amazon*
Micro HDMI to HDMI cables 2-pack	1	~$10	*Amazon*

Radio (LoRa)

Item	Qty	Est. Price	Where
Heltec WiFi LoRa 32 V3 915MHz	2	~$20 ea	*heltec.org / Rokland*
U.FL–to–SMA pigtail 6” (IPEX)	2	~$5 ea	*Amazon*
915 MHz LoRa antenna (SMA-F)	2	~$8 ea	*Amazon*

Category	Est. Subtotal
Compute & Storage	~$396
Rack & Networking	~$342
Power	~$190
Radio	~$66
TOTAL	**~$994**

Build Phases

Full step-by-step instructions are in M2_BUILD_GUIDE.md. This is the phase summary — what happens in what order and roughly how long it takes.

Phase	What Happens	Est. Time
0 — Pre-Build	Monero pre-sync on x86 PC. Software downloads. microSD prep.	*1–2 days*
1 — Rack Assembly	Physical rack build. Cable management. Component mounting.	*1–2 hours*
2 — Network Baseline	GL.iNet WiFi config. DHCP. Static IPs for both Pis. DNS overrides.	*30 min*
3 — OS & Base Config	Raspberry Pi OS Lite on both Pis. Docker. SSH hardening. NVMe boot.	*1–2 hours*
4 — Pi #1 Comms Stack	Conduit, Element Web, Nginx, i2pd, Tor, Cloudflared, AdGuard Home.	*1–2 hours*
5 — Pi #2 Tactical Stack	OpenTAK Server (native), Monero rsync, Reticulum, Headscale, Mumble.	*1–2 hours*
6 — LoRa Radio Setup	Flash LEFT Heltec with RNode firmware. Flash RIGHT Heltec with Meshtastic.	*30 min*
7 — Integration Testing	ATAK connect. GeoChat. Position sharing. Matrix federation check. Mesh verify.	*30–60 min*
8 — Field Hardening	Firewall rules. Backup. Credential rotation. Field card print. Kiosk setup.	*30 min*

Timeline Reality Check

- Monero sync is the long pole. Do it first, in parallel, on any spare x86 PC. The Pi 5 cannot sync Monero from scratch in a reasonable timeframe — always rsync from a pre-synced machine.
- Phases 1–8 take one focused day. With the Monero DB ready, the full rack build and software stack is a one-day project for someone comfortable with Linux and SSH.
- Phases 4–5 are the hardest. OpenTAK Server installs its own PostgreSQL and RabbitMQ — follow the official OTS install script exactly. Do not use Docker for OTS.

Software Stack

All open-source. All self-hosted. No SaaS, no subscriptions.

Service	Node	What It Does	Runtime
Conduit	Pi #1	Matrix homeserver — encrypted group chat	*Docker*
Element Web	Pi #1	Browser-based Matrix client	*Docker*
Nginx	Pi #1	Reverse proxy — routes all Pi #1 traffic	*Docker*
Tor	Pi #1	Hidden service (.onion) for Matrix + community page	*Docker*
i2pd	Pi #1	I2P eepsite — community page on I2P	*Docker*
Cloudflared	Pi #1	Clearnet tunnel — exposes Matrix/Element/OTS to internet	*Docker*
AdGuard Home	Pi #1	DNS resolver + ad blocking — also bypasses Android Private DNS	*Docker*
OpenTAK Server	Pi #2	TAK CoT relay server — ATAK connect, positions, GeoChat, web map	*Systemd (native)*
Mumble	Pi #2	Push-to-talk voice (use Mumla on Android)	*Docker*
Reticulum + NomadNet	Pi #2	Encrypted mesh transport + LoRa BBS	*Python service*
Monerod	Pi #2	Pruned Monero node — community privacy coin endpoint	*Docker*
Headscale	Pi #2	Self-hosted WireGuard VPN coordination — remote ATAK access	*Docker*
Mosquitto	Pi #2	MQTT broker for IoT/sensor integration	*Docker*

Operating System

Raspberry Pi OS Lite (64-bit, Bookworm) on both Pis. No desktop environment — headless SSH only. Docker Engine (not Docker Desktop) manages all containerized services.

Skills You'll Need

Skill	Required For	Level
Linux command line (SSH)	Everything — all config is CLI-based	*Comfortable*
Docker Compose	Pi #1 full stack + most Pi #2 services	*Basic*
Networking basics	IP assignment, DNS, firewall rules	*Basic*
Systemd	OpenTAK Server management and troubleshooting	*Basic*
Git + file transfer (SCP)	Config deployment, script updates	*Basic*
ATAK app familiarity	Integration testing (Phase 7)	*Helpful*

Key Gotchas — Read These Before You Start

- Monero MUST be pre-synced on an x86 machine. Syncing natively on ARM Pi takes weeks. Sync on a spare laptop or desktop, then rsync the blockchain directory to Pi #2.
- OpenTAK Server is NOT Docker. It installs its own PostgreSQL and RabbitMQ via the official install script. Do not try to containerize it — use the native systemd install.
- Both Heltec V3 LoRa boards are identical hardware. The only difference is firmware: LEFT panel position gets RNode (Reticulum), RIGHT gets Meshtastic. Label them before flashing.
- GL.iNet DNS overrides are required. Set static DNS entries on the GL.iNet router pointing matrix.capableenough.org and atakenroll.capableenough.org to the correct Pi IPs. Without this, Cloudflare-tunneled domains won't resolve on the local LAN.
- NVMe must be M.2 2230 form factor. Standard 2280 drives will not fit in the GeeekPi dual mount. Verify the 2230 suffix before ordering — prices are volatile (Steam Deck demand).
- Android phones block local DNS by default. The AdGuard Home container on Pi #1 handles DNS resolution for the LAN. GL.iNet DHCP serves Pi #1 (192.168.8.10) as the DNS server.
- ATAK needs 'Consistent' reporting. Stationary phones default to event-driven reporting and won't appear on the map. Tell all users: Settings → Reporting → Consistent.

Full Documentation

Resource	Contents
M2_BUILD_GUIDE.md	Complete step-by-step build (all phases, all commands)
M2_Community_Node_Runbook.pdf	Field operations guide — power-on, onboarding, troubleshooting
M2_MiniRack_Hardware_BOM.md	Full BOM with ASINs, verified pricing, rack layout details
M2_ATAK_Connectivity_Guide.md	ATAK onboarding deep dive — all three connection paths
M2_MASTER_COMMUNITY_NODE_PLAN.md	Architecture, service inventory, design rationale
github.com/PapaSierra555/M2-Community-Node	All source files, scripts, and generators